# The crystal has disappeared from view

**Learning intention**

**Describe melting, freezing and dissolving and explain how the starting substances can be recovered in these examples.**

Did the substance vanish, melt or dissolve?

A detective needs more than appearance.

# State or mixture?

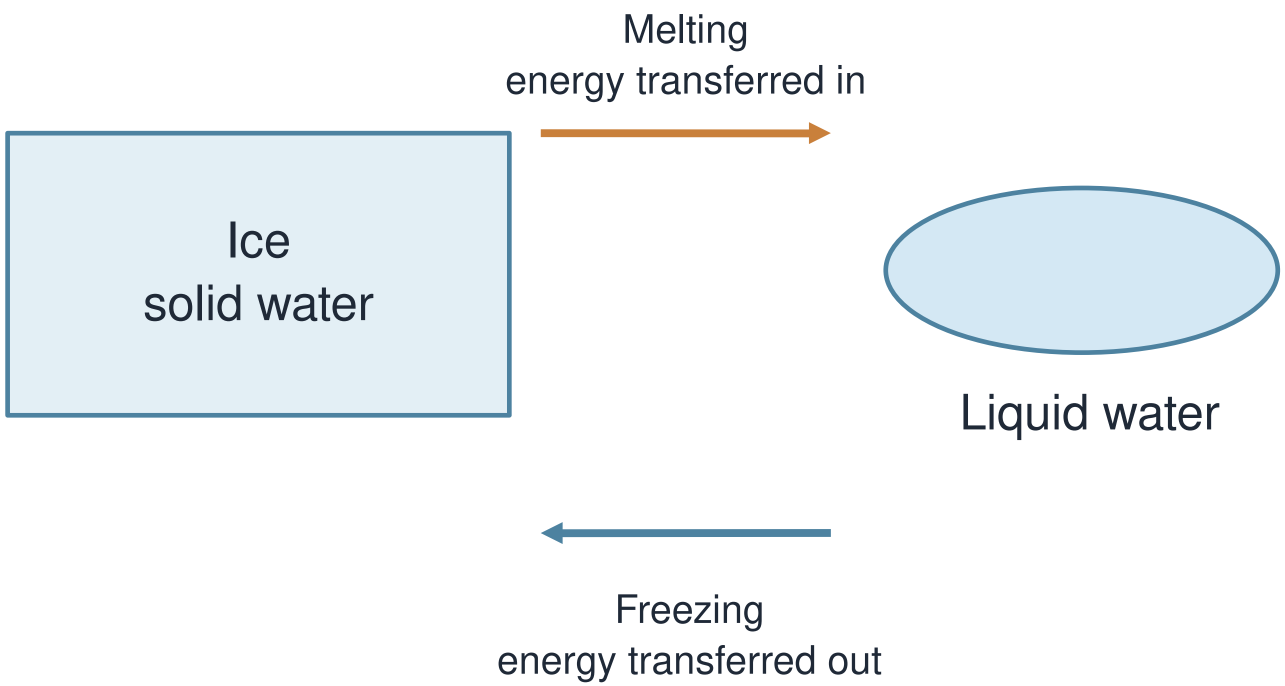
R1. Name a solid and a liquid.

R2. Does invisible always mean absent?

R3. What unit records mass?

# I identify a change of state

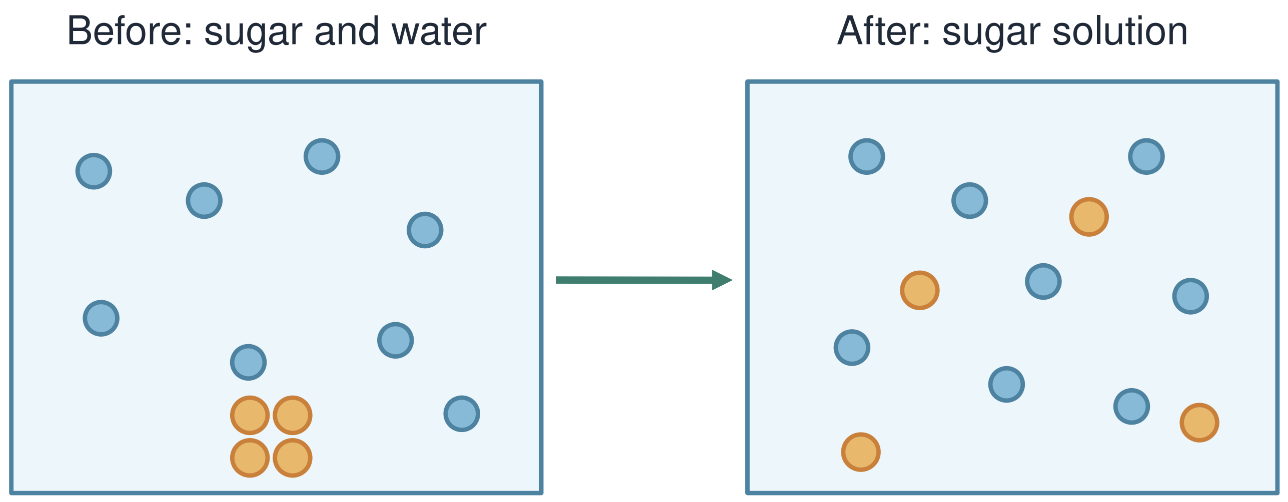
### Same substance; different state




Ice and liquid water are the same substance. Melting changes solid to liquid. Freezing changes liquid to solid.

# I count what the model retains

**Dissolving: sugar remains in the mixture**




Blue: water molecules. Orange: sugar molecules.
Symbols and quantities are not to scale.

Sugar molecules spread among water molecules. The sugar symbols remain present. A solution forms; the sugar has not melted here.

# I choose a route back

### Recovering the dissolved solid

Sugar solution → Water evaporates over time → Sugar remains after drying

This recovers the sugar. Water is not collected
by simple open evaporation; no heating is required here.

Letting water evaporate can leave sugar behind. Simple open evaporation does not collect the water. Reversible does not mean instant or effortless.

# Process evidence tribunal

W1. Classify A and B: melting/freezing or dissolving. W2. Use C and D to challenge “the sugar vanished”. State what each piece of evidence contributes.

| Constructed record | Before | After |
|---|---|---|
| Sealed mass / g | 120 | 120 |
| Open recovery over days | Sugar solution | Dry sugar remains |

# Choose a reverse process

A tray contains melted ice. A second contains sugar solution.

Which needs enough cooling? Which needs water removed to recover sugar?

# Write a precise verdict

W3. Repair “Dissolving makes sugar into water”.

Use the particle model and recovery record.

# Sugar is stirred into room-temperature water

The crystals become invisible and a solution forms. What happened?

**A** The sugar dissolved and remains in the solution.

**B** The sugar melted into pure liquid sugar.

**C** The sugar stopped existing.

# Your independent investigation

Choose one response route.

Compare melting, freezing and dissolving.

Explain a route to recover the stated substance.

# Use the evidence

Track the substance before and after.

Use mass and recovery evidence appropriately.

Name what a chosen process actually collects.

# Check before sharing

Does my explanation answer the question?

Have I named the evidence or process?

Have I kept the conclusion within its limits?

# Compare and improve

Share one corrected process label.

Check the before-and-after substance.

Improve a disappearing or melting claim.

# A question worth investigating

Suggest one next question about this lesson’s evidence.

Choose a question whose answer would improve our model.

# Show the science you can explain

E1. How is melting different from dissolving?

E2. Where is the sugar in a clear solution?

E3. How could sugar be recovered here?